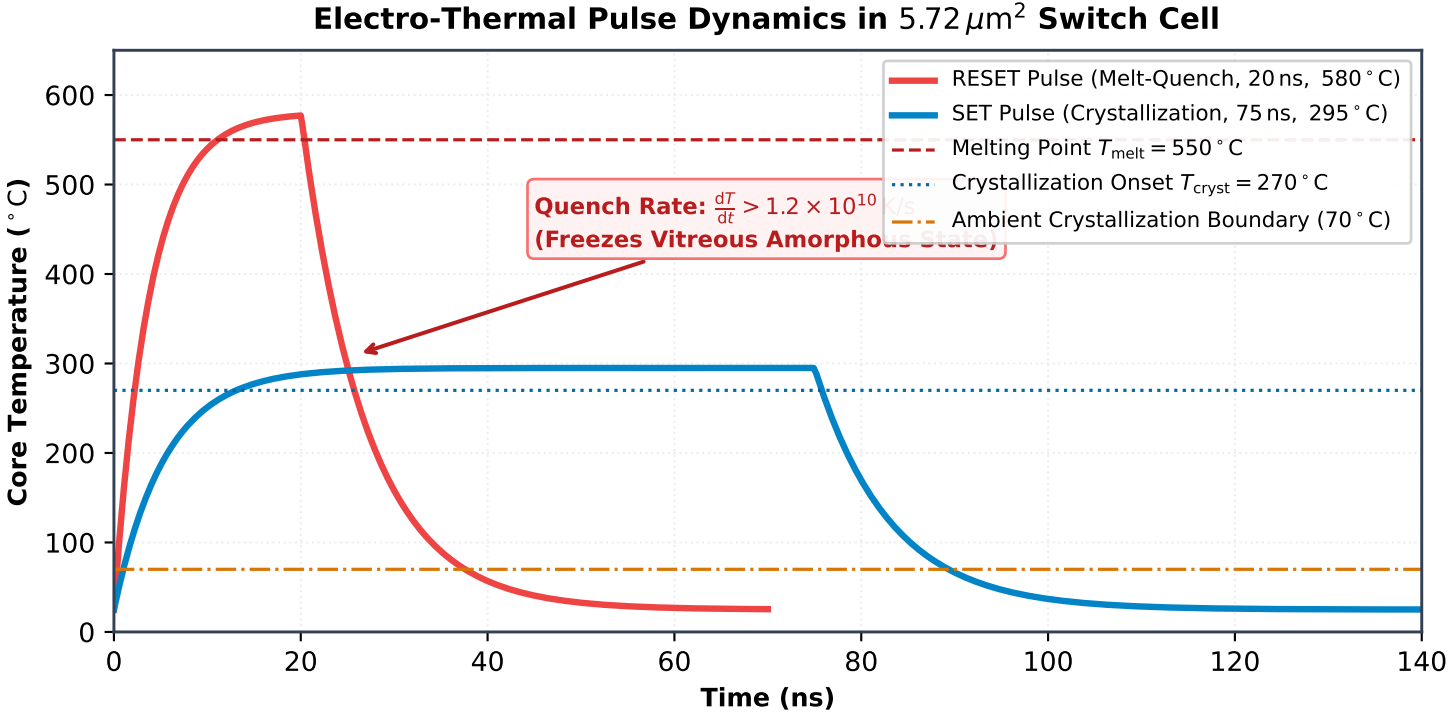
Electro-Thermal Pulse Dynamics in 5.72 μm^2 Switch Cell
RESET Pulse (Melt-Quench, 20 ns, 580 °C)
SET Pulse (Crystallization, 75 ns, 295 °C)
Melting Point $T_{melt} = 550$ °C
Crystallization Onset $T_{cryst} = 270$ °C
Ambient Crystallization Boundary (70 °C)
Quench Rate: $\frac{dT}{dt} > 1.2 \times 10^{10}$ K/s
(Freezes Vitreous Amorphous State)
Core Temperature (°C)
Time (ns)
0
100
200
300
400
500
600
0
20
40
60
80
100
120
140